KO LIH HAN

Software Engineer | Applied AI & LLM Systems

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EXPERIENCE

Software Developer (Part-time)

Kahap Inc | Aug 2022 – Aug 2026

- Owned the 0-to-1 build and production rollout of a feedback platform, centralizing submissions, discussions, admin review, and notifications into one operational system.
- Operated a Laravel/Vue.js career platform across queue workers, scheduled jobs, SSL renewals, database/server migrations, and Linux hosting.
- Integrated GPT-SoVITS plus automated pronunciation and text checks into existing backend workflows for AI-assisted content features.

Full-Stack Developer Intern

Pac Com Solution | Jun 2021 – Jan 2022

- Shipped and debugged Laravel/Vue.js features across backend and frontend paths in a production POS codebase.

PROJECTS

Small-Agent QLoRA

2026 | github.com/kolihhan/small-agent-qlora

Python, Qwen3.5-9B, QLoRA / PEFT, Transformers, llama.cpp

- Built a single-model tool agent for search, file inspection, and Python execution, with bounded tool errors, duplicate-call blocking, timeouts, source-size limits, and persisted traces.
- Diagnosed policy failures from traces and turned them into state-aware corrective QLoRA supervision, raising exact closed-loop success from 34.5% to 62.0% on 200 frozen tasks.
- Rejected full-trajectory SFT after held-out regression; on GAIA100, an intermediate tuned checkpoint doubled base accuracy from 9% to 18%, while the final continuation held 18% but exposed over-search / failure-to-finalize.

Semantic Text-to-SQL

2026 | github.com/kolihhan/semantic-text2sql-agent

Python, FastAPI, LangGraph, SQLite, Qwen3.5-9B

- Designed a local NL-to-SQL service around a key reliability gap: executable SQL can still be wrong; combined schema grounding, deterministic preflight, bounded repair, read-only execution, and explicit traces.
- On 100 frozen BIRD DEV queries, guarded verification/repair raised conservative execution success from 75% to 90% and EX-compatible match from 33% to 39%, recovering 6 wrong answers with 0 regressions.
- Kept the simpler guarded path after semantic-review and grounding ablations failed to improve net correctness; reused one backend layer across FastAPI, CLI, and Streamlit.

RESEARCH

Context-Aware Implicit Workplace Meaning

M.S. Thesis | National Tsing Hua University

NLP, retrieval, role/context knowledge graphs, Llama 3.1-8B

- Built a listener-side NLP pipeline for implicit workplace meaning, curating 20,455 utterances across 7 boards and labeling speaker/listener state, emotion, and communicative purpose.
- Built separate Role KG and Context KG retrieval paths and fused top-1 evidence into local Llama 3.1-8B generation to model who said what, to whom, and under what dialogue context.
- Designed a controlled evaluation on the same 400 aligned targets with fixed prompts across target-only and preceding-context arms, isolating context/evidence changes from prompt drift.
- Dual-KG improved target-only semantic F1 from 0.644 to 0.682 and preceding-context F1 from 0.680 to 0.691; the baseline context gain (+0.036) showed dialogue history remained a major signal alongside graph evidence.

SKILLS

AI / LLM: Agents & tool use, RAG / retrieval, model evaluation, PEFT / QLoRA, Hugging Face Transformers, LangGraph, llama.cpp

Backend: Python, FastAPI, PHP, Laravel, SQL, REST APIs, JavaScript

Data / Infrastructure: MySQL, SQLite, Linux, Git, GitHub Actions, pytest

Frontend: Vue.js, React

Languages: Chinese (Native/Bilingual), English (Fluent), Malay (Intermediate)

EDUCATION

National Tsing Hua University | Master of Science, Information Systems & Applications | GPA: 4.09/4.3 | Jun 2026

Chung Hua University | Bachelor of Science, Computer Science & Information Engineering | GPA: 3.9/4.0 | Jun 2024